

Practical No.12

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25-10-2025

Objective:

Create a dataset and compute following

1. Group data by Region and compute summary statistics of Revenue
2. Perform multiple aggregations per region by computing (mean, sum and max) for Units_sold and (sum and mean) for Revenue
3. Apply a custom function to each group for calculating revenue per unit sold.
4. Use map function to assign Zone1 to East, Zone-2 to West , Zone3 to South and Zone4 to North.
5. Calculate normalized revenue within each region
6. Compute z-score of Revenue within each region.

Python Codes:

Creation of Dataframe

```
In [2]: import pandas as pd
import numpy as np

# Sample data
data = {
    'Region': ['East', 'West', 'East', 'South', 'West', 'East', 'South', 'West', 'East', 'South'],
    'Salesperson': ['Amit', 'Brian', 'Cathy', 'Deepa', 'Eva', 'Farah', 'George', 'Himanshu', 'Isha', 'Jatin'],
    'Product': ['Laptop', 'Laptop', 'Tablet', 'Tablet', 'Laptop', 'Tablet', 'Tablet', 'Laptop', 'Laptop', 'Tablet'],
    'Units_Sold': [15, 10, 8, 12, 9, 14, 7, 13, 11, 9],
    'Revenue': [150000, 100000, 64000, 96000, 90000, 112000, 56000, 130000, 110000, 72000]
}

df = pd.DataFrame(data)
df
```

```
Out[2]:
```

	Region	Salesperson	Product	Units_Sold	Revenue
0	East	Amit	Laptop	15	150000
1	West	Brian	Laptop	10	100000
2	East	Cathy	Tablet	8	64000
3	South	Deepa	Tablet	12	96000
4	West	Eva	Laptop	9	90000
5	East	Farah	Tablet	14	112000
6	South	George	Tablet	7	56000
7	West	Himanshu	Laptop	13	130000
8	East	Isha	Laptop	11	110000
9	South	Jatin	Tablet	9	72000

1. Group data by Region and compute summary statistics of Revenue

```
In [3]: grouped = df.groupby('Region')
region_revenue = grouped['Revenue'].describe()
print(region_revenue)
```

	count	mean	std	min	25%	50%	\
Region							
East	4.0	109000.000000	35194.696570	64000.0	98500.0	111000.0	
South	3.0	74666.666667	20132.891827	56000.0	64000.0	72000.0	
West	3.0	106666.666667	20816.659995	90000.0	95000.0	100000.0	
	75%	max					
Region							
East	121500.0	150000.0					
South	84000.0	96000.0					
West	115000.0	130000.0					

2. Perform multiple aggregations per region by computing (mean, sum and max) for Units_sold and (sum and mean) for Revenue

```
In [4]: # Aggregate multiple statistics
agg_results = df.groupby('Region').agg({
    'Units_Sold': ['mean', 'sum', 'max'],
    'Revenue': ['sum', 'mean']
})
agg_results
```

```
Out[4]:
```

	Units_Sold			Revenue	
	mean	sum	max	sum	mean
Region					
East	12.000000	48	15	436000	109000.000000
South	9.333333	28	12	224000	74666.666667
West	10.666667	32	13	320000	106666.666667

3. Apply a custom function to each group for calculating revenue per unit sold.

```
In [6]: # Custom function to compute Revenue per Unit within each group
import warnings
warnings.filterwarnings("ignore", category=DeprecationWarning)
def revenue_per_unit(group):
    group['Revenue_per_Unit'] = group['Revenue'] / group['Units_Sold']
    return group

applied_df = df.groupby('Region').apply(revenue_per_unit)
applied_df
```

```
Out[6]:
```

	Region	Salesperson	Product	Units_Sold	Revenue	Revenue_per_Unit
Region						
East	0	East	Amit	Laptop	15	150000
	2	East	Cathy	Tablet	8	64000
	5	East	Farah	Tablet	14	112000
	8	East	Isha	Laptop	11	110000
South	3	South	Deepa	Tablet	12	96000
	6	South	George	Tablet	7	56000
	9	South	Jatin	Tablet	9	72000
West	1	West	Brian	Laptop	10	100000
	4	West	Eva	Laptop	9	90000
	7	West	Himanshu	Laptop	13	130000

4. Use map function to assign Zone1 to East, Zone-2 to West , Zone3 to South and Zone4 to North.

```
In [7]: region_map = {'East': 'Zone 1', 'West': 'Zone 2', 'South': 'Zone 3'}

# Map new Labels to regions
df['Zone'] = df['Region'].map(region_map)
df
```

```
Out[7]:
```

	Region	Salesperson	Product	Units_Sold	Revenue	Zone
0	East	Amit	Laptop	15	150000	Zone 1
1	West	Brian	Laptop	10	100000	Zone 2
2	East	Cathy	Tablet	8	64000	Zone 1
3	South	Deepa	Tablet	12	96000	Zone 3
4	West	Eva	Laptop	9	90000	Zone 2
5	East	Farah	Tablet	14	112000	Zone 1
6	South	George	Tablet	7	56000	Zone 3
7	West	Himanshu	Laptop	13	130000	Zone 2
8	East	Isha	Laptop	11	110000	Zone 1
9	South	Jatin	Tablet	9	72000	Zone 3

5. Calculate normalized revenue within each region

```
In [8]: df['Revenue_Normalized'] = df.groupby('Region')['Revenue'].transform(lambda x: x / x.mean())
df[['Region', 'Revenue', 'Revenue_Normalized']]
```

```
Out[8]:
```

	Region	Revenue	Revenue_Normalized
0	East	150000	1.376147
1	West	100000	0.937500
2	East	64000	0.587156
3	South	96000	1.285714
4	West	90000	0.843750
5	East	112000	1.027523
6	South	56000	0.750000
7	West	130000	1.218750
8	East	110000	1.009174
9	South	72000	0.964286

6. Compute z-score of Revenue within each region.

```
In [9]: df['Revenue_Zscore'] = df.groupby('Region')['Revenue'].transform(lambda x: (x - x.mean()) / x.std())
df[['Region', 'Salesperson', 'Revenue', 'Revenue_Zscore']]
```

Out[9]:

	Region	Salesperson	Revenue	Revenue_Zscore
0	East	Amit	150000	1.164948
1	West	Brian	100000	-0.320256
2	East	Cathy	64000	-1.278602
3	South	Deepa	96000	1.059626
4	West	Eva	90000	-0.800641
5	East	Farah	112000	0.085240
6	South	George	56000	-0.927173
7	West	Himanshu	130000	1.120897
8	East	Isha	110000	0.028413
9	South	Jatin	72000	-0.132453

Interpretation:

1. Here we grouped data by Region and computed summary statistics of Revenue.
2. Performed multiple aggregations per region by computing (mean, sum and max) for Units_sold and (sum and mean) for Revenue
3. We Apply a custom function to each group for calculating revenue per unit sold.
4. Usnig map function we assigned Zone1 to East, Zone-2 to West , Zone3 to South and Zone4 to North.
5. Calculated normalized revenue within each region.
6. Computed z-score of Revenue within each region.